

PaddyGuard AI: A Multimodal AI Based System for Rice Leaf Disease & Pest Detection, Treatment Recommendation and Knowledge Assistance

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Rice is the primary staple crop for over 21 million people in Sri Lanka, occupying approximately 34% of the nation's agricultural land. Despite its central importance, rice cultivation is severely threatened by a range of diseases including Bacterial Blight (*Xanthomonas oryzae* pv. *oryzae*), Rice Blast (*Magnaporthe oryzae*), and Brown Spot (*Bipolaris oryzae*), collectively responsible for yield losses between 30% and 80% if left undetected. Sri Lankan paddy farmers, many of whom are semi-literate and operate in rural settings with limited access to agronomists, are particularly vulnerable to delayed or inaccurate disease diagnosis.

Existing AI-based plant disease detection systems predominantly rely on image inputs, requiring smartphone cameras and sufficient lighting conditions that are often unavailable in field settings. Furthermore, virtually all such systems operate exclusively in English, creating a significant accessibility barrier for Sinhala-speaking farming communities. This creates a critical research gap: no existing solution enables low-literacy Sri Lankan farmers to describe symptoms in their native Sinhala language via voice and receive an immediate, actionable disease diagnosis.

This research proposes the development of the voice-based symptom understanding and disease classification component of PaddyGuard AI - a multimodal, mobile-responsive web application for rice disease and pest management. The proposed component allows farmers to narrate observed symptoms in Sinhala (e.g. ‘ගොයම් පැලවල පොඩි කළු ලප තියේ’ - meaning ‘Paddy plant leaves have small black spots’), which are processed through an Automatic Speech Recognition (ASR) pipeline using OpenAI Whisper, followed by NLP-based symptom extraction and classification using transformer models (BERT/RoBERTa) to predict the disease. The system further engages farmers in adaptive voice-based follow-up questions when image evidence is unavailable and seamlessly integrates with image-based disease detection and pest identification modules when images are uploaded. A text-to-speech (TTS) module returns all predictions and recommendations in spoken Sinhala.

The expected outcome is a voice-first, language-inclusive diagnostic pipeline achieving a classification accuracy of no less than 85% across four disease categories. The system is projected to serve over two million rice-farming households in Sri Lanka, contributing directly to SDG 2 (Zero Hunger) and SDG 9 (Industry, Innovation, and Infrastructure).

Keywords: Automatic Speech Recognition; Natural Language Processing; Rice Disease Classification; Sinhala Language Processing; Voice-Based Agricultural Advisory; Transformer Models; Multimodal AI.

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TABLE OF CONTENT

Contents

ABSTRACT.....	3
ACKNOWLEDGEMENT	5
TABLE OF CONTENT	6
LIST OF TABLES	8
LIST OF FIGURES	9
LIST OF ABBREVIATIONS	10
1. INTRODUCTION	11
1.1 Background and Context.....	11
1.2 Literature Review.....	12
1.3 Research Gap	15
2. OBJECTIVE	16
2.1 Main Objective.....	16
2.2 Specific Objectives	16
3. METHODOLOGY	17
3.1 Research Approach	17
3.2 Data Collection Strategy	17
3.3 Technical Implementation Pipeline.....	18
3.4 Comparison of Existing Systems	19
3.5 Validation Strategy.....	20
4. HIGH-LEVEL SYSTEM ARCHITECTURE.....	21
4.1 Overall PaddyGuard AI System Overview	21
4.2 Detailed Architecture (Voice-NLP Pipeline).....	23
4.4 Data Flow Diagram.....	24
4.5 Scalability and Security Considerations	24
5. USER REQUIREMENTS	25
5.1 Stakeholder Identification.....	25
5.2 Functional Requirements	25
5.3 Non-Functional Requirements	26

6. COMMERCIALIZATION PLAN	27
6.1 Target Market and Customer Persona	27
6.2 Value Proposition	27
6.3 Revenue Model	27
6.4 Competitive Advantage.....	28
6.5 Intellectual Property	28
7. BUDGET AND JUSTIFICATION	29
7.1 Estimated Project Budget.....	29
7.2 Hardware Costs	29
7.3 Software Licenses	30
7.4 Cloud Services	30
7.5 Human Effort Cost.....	31
7.6 Justification of Expenses.....	32
8. WORK BREAKDOWN STRUCTURE (WBS).....	33
9. GANTT CHART.....	34
10. References List.....	35
11. APPENDICES.....	37
11.1 Preliminary UI mockups	37

LIST OF TABLES

Table 1: List of Abbreviations.....	10
Table 2: Comparison of Existing System.....	19
Table 3: NLP pipeline stages.....	23
Table 4: Hardware cost	29
Table 5:Software License.....	30
Table 6: Cloud Service.....	31
Table 7: Human Effort Cost.....	32
Table 8: Justification of Expenses.....	32

LIST OF FIGURES

Figure 1:High Level Digram.....	22
Figure 2: Voice Based Rise Disease Diagnosis Component High-Level Digram	23
Figure 3: Voice Based Rise Disease Diagnosis Component Data Flow	24
Figure 4: WBS	33
Figure 5: Gantt Chart	34
Figure 6: Leafe Diagnosis	37
Figure 7: Home Page	37
Figure 8: Chat Bot.....	38
Figure 9: Pets Detection.....	38
Figure 10: Result Page.....	39
Figure 11: Voice Diagnosis	39

LIST OF ABBREVIATIONS

Table 1: List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
DL	Deep Learning
FAISS	Facebook AI Similarity Search
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
OOD	Out-of-Distribution
RAG	Retrieval-Augmented Generation
RoBERTa	Robustly Optimized BERT Pretraining Approach
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SVM	Support Vector Machine
TF-IDF	Term Frequency – Inverse Document Frequency
TTS	Text-to-Speech
UI	User Interface
WBS	Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

Rice cultivation forms the backbone of Sri Lanka's agricultural economy and food security. According to the Food and Agriculture Organization (FAO), Sri Lanka produced approximately 2.7 million metric tons of paddy in 2022, with rice being cultivated across over 900,000 hectares [1]. The farming population, estimated at over 1.8 million households, is primarily concentrated in rural districts including Polonnaruwa, Kurunegala, Anuradhapura, and Ampara, where access to professional agricultural advisory services is severely limited.

Rice crops in Sri Lanka are persistently vulnerable to three major diseases: (1) Bacterial Blight, caused by *Xanthomonas oryzae* pv. *oryzae*, produces water-soaked lesions along leaf margins that gradually turn yellow to white, capable of reducing yield by up to 50%; (2) Rice Blast (Leaf Blast), caused by the fungal pathogen *Magnaporthe oryzae*, manifests as characteristic diamond-shaped grey lesions with brown borders and is considered the single most devastating rice disease globally; and (3) Brown Spot, caused by *Bipolaris oryzaicola*, produces circular to oval dark brown lesions and thrives under nutrient-deficient conditions, often reducing grain yield by 45% in severe cases [2]. Early and accurate identification of these diseases is critical for timely intervention.

Despite growing adoption of digital technologies, a significant digital divide persists among Sri Lankan paddy farmers. A 2021 survey by the Hector Kobbekaduwa Agrarian Research and Training Institute found that while mobile phone penetration is high (approximately 78% among rural farming communities), smartphone ownership with camera capabilities remains below 35% in many rice-growing districts. Moreover, adult literacy rates in rural agricultural zones remain around 70%, and Sinhala is the exclusive language of communication for the vast majority of farmers [3].

Existing AI-powered agricultural disease detection systems, while technically sophisticated, are almost entirely designed around image-based inputs and English-language interfaces. Systems such as PlantVillage by Mohanty et al. [4] demonstrated that convolutional neural networks (CNNs) can achieve over 99% accuracy on laboratory-quality leaf images; however, their performance degrades substantially under real-world field conditions with variable lighting, angles, and image quality. More critically, they offer no pathway for farmers who cannot operate cameras or who wish to describe symptoms verbally.

PaddyGuard AI addresses this critical accessibility gap through a multimodal, mobile-responsive web application. The system’s architecture is built around four integrated components: (1) Voice-based symptom understanding and NLP classification (this research), (2) Image-based rice leaf disease classification with Out-of-Distribution (OOD) detection, (3) Hybrid AI and knowledge-based treatment recommendation with an RAG chatbot, and (4) Rice pest detection using deep learning with OOD detection. Together, these components create a holistic, inclusive diagnostic and advisory platform accessible to low-literacy Sinhala-speaking farmers.

This research component specifically focuses on enabling farmers to describe what they observe in their paddy fields using natural Sinhala speech (e.g., ‘මගේ ගොයම් කොළ කහපටවෙලා’ – ‘My paddy leaves have turned yellow’, or ‘කොළ මත කළු පැල්ලම් තිබේ’ – ‘There are black spots on the leaves’), convert that speech to text via ASR, extract symptom-relevant entities through NLP, classify the disease, and return both text and synthesized speech responses. When uncertainty is high, the system initiates an intelligent adaptive dialogue with follow-up questions before escalating to image-based verification if available.

This research aligns with UN Sustainable Development Goal 2 (Zero Hunger) by improving crop yield protection through accessible disease management tools, and with SDG 9 (Industry, Innovation and Infrastructure) by advancing inclusive and sustainable technological innovation for agriculture in developing nations.

1.2 Literature Review

The intersection of speech technology, NLP, and agricultural advisory systems is a relatively nascent but rapidly growing research domain. This section reviews the most relevant recent scholarly contributions across the key technical layers of the proposed system.

1.2.1 Automatic Speech Recognition (ASR) for Low-Resource Languages

Radford et al. introduced Whisper, a large-scale weakly supervised speech recognition model trained on 680,000 hours of multilingual audio data [5]. Whisper demonstrates near state-of-the-art performance across multiple languages including Sinhala, making it a compelling backbone for the ASR component of this research. Critically, Whisper’s multilingual capability enables direct transcription of Sinhala spoken input without requiring separate language identification steps. Prior to Whisper, Sinhala ASR was largely reliant on limited, domain-specific datasets; Povey et

al.'s Kaldi toolkit [6] has been used in academic Sinhala ASR research but with substantially lower accuracy on spontaneous agricultural speech compared to transformer-based approaches.

1.2.2 NLP-Based Text Classification for Agricultural Symptom Understanding

Devlin et al.'s BERT [7] established a new paradigm for NLP by introducing bidirectional transformer pre-training that captures deep contextual relationships within text. For short, conversational symptom descriptions - such as those a farmer might speak – BERT's contextual embeddings significantly outperform traditional bag-of-words models. Liu et al.'s RoBERTa [8], a robustly optimized variant, further improves classification performance by removing the next-sentence prediction objective and training with larger batches and more data. Both models can be fine-tuned on domain-specific agricultural symptom corpora to produce high-accuracy disease classifiers.

Traditional ML approaches remain competitive baselines. TF-IDF vectorization combined with Support Vector Machines (SVM) has been widely applied to agricultural text classification tasks, with Sethy et al. [9] demonstrating SVM-based rice disease prediction from textual descriptions achieving over 82% accuracy. However, transformer models consistently outperform TF-IDF+SVM under noisy, informal natural language conditions typical of voice-transcribed farmer speech.

1.2.3 Voice-Based Agricultural Advisory Systems

The deployment of voice-based interfaces in agricultural contexts in developing nations has been explored in several pilot studies. IVR (Interactive Voice Response) systems have been deployed in sub-Saharan Africa and South Asia to deliver crop advisory services to smallholder farmers, with studies demonstrating uptake rates of 60–80% among previously underserved communities [10]. However, these systems are largely rule-based and do not incorporate AI-based symptom classification. More recently, the integration of conversational AI with agricultural advisory represents a promising frontier; however, Sinhala-language voice-first diagnostic systems for rice diseases remain entirely unaddressed in the current literature.

1.2.4 Text-to-Speech for Regional Language Accessibility

Modern neural TTS systems, including those based on the Tacotron architecture and VITS (Variational Inference with adversarial learning for end-to-end TTS), produce natural-sounding speech across multiple languages. Google Cloud TTS and AWS Polly both support Sinhala language synthesis as of 2024, offering practical deployment options for returning diagnostic responses as synthesized speech - a critical feature for low-literacy users.

1.2.5 Multimodal Agricultural AI Systems

Recent research has demonstrated the superiority of multimodal approaches that combine visual and language modalities for agricultural decision support. Mohanty et al.'s foundational PlantVillage work [4] established the viability of image-based plant disease detection. Subsequent research by Singh and Misra [11] extended this to field conditions with varying lighting. The integration of voice input with image-based confirmation, as proposed in this research, represents a novel multimodal architecture that has not been previously explored for rice disease diagnosis in South Asian agricultural contexts.

1.2.6 Adaptive Dialogue Systems for Domain-Specific Q&A

Adaptive dialogue systems that ask follow-up questions to reduce uncertainty in diagnosis share conceptual foundations with medical symptom checkers such as Babylon Health and Ada Health. These systems employ decision-tree or probabilistic frameworks to guide users toward progressively refined diagnoses. Howard and Ruder [12] demonstrated that language model pre-training followed by domain-specific fine-tuning enables effective transfer to specialized classification tasks with limited labelled data - directly relevant to the challenge of building a Sinhala agricultural symptom corpus from scratch.

1.3 Research Gap

A systematic analysis of the existing literature and commercial systems reveals the following specific gaps that this research component addresses:

- **Language exclusivity:** All existing AI-based rice disease detection and advisory systems are designed exclusively for English or high-resource Asian languages (Mandarin, Hindi). No published system supports Sinhala voice input for rice disease diagnosis, leaving Sri Lanka's largest farming demographic entirely underserved.
- **Modality dependency on images:** Existing precision agriculture AI systems require visual input (leaf photographs) as their primary modality. No system provides a voice-only fallback diagnostic pathway that is functionally complete - i.e., capable of reaching a confident disease classification through voice dialogue alone when images are unavailable.
- **Absence of multimodal orchestration:** No existing system dynamically orchestrates between voice-first questioning and image-based confirmation in a unified user flow. The interplay between ASR-driven NLP classification and CNN-based image classification has not been architected as a coherent diagnostic pipeline.
- **Lack of OOD-awareness in NLP classification:** Prior NLP-based disease classification systems do not incorporate mechanisms to detect when a farmer's description does not correspond to any of the trained disease categories (OOD scenario), risking confident misclassification of novel or unclear symptoms.
- **No bidirectional voice accessibility:** Existing systems return text-only outputs or require users to read predictions on screen - a critical barrier for low-literacy farmers. A complete Sinhala TTS output layer for returning diagnostic results as natural speech is absent from all reviewed systems.

This research addresses all five gaps by proposing the first Sinhala voice-enabled, multimodal, OOD-aware rice disease classification component with full text-to-speech response capability, integrated within the PaddyGuard AI ecosystem.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a Sinhala voice-based symptom understanding and NLP-driven rice disease classification system that enables low-literacy Sri Lankan paddy farmers to receive accurate, speech-delivered disease diagnoses and seamlessly integrates with the image-based disease detection, pest detection, and treatment recommendation components of the PaddyGuard AI platform.

2.2 Specific Objectives

1. To build and curate a Sinhala agricultural symptom dataset containing a minimum of 1,500 labelled voice utterances covering Bacterial Blight, Rice Blast, Brown Spot, and Healthy plant categories, sufficient for training and evaluating ASR-NLP classification pipelines.
2. To implement and fine-tune an Automatic Speech Recognition pipeline using OpenAI Whisper on Sinhala agricultural speech, achieving a Word Error Rate (WER) of no greater than 15% on domain-specific test utterances.
3. To develop and compare NLP classification models - including TF-IDF+SVM (baseline), fine-tuned BERT, and fine-tuned RoBERTa - and identify the optimal model achieving classification accuracy of at least 85% on the four disease/health categories.
4. To design and implement an adaptive voice-based follow-up questioning module that reduces diagnostic uncertainty when initial confidence scores fall below a defined threshold (0.70), using a decision-tree-guided dialogue framework.
5. To integrate a Text-to-Speech (TTS) output module that converts disease predictions and follow-up questions into natural Sinhala speech, achieving a Mean Opinion Score (MOS) of at least 3.5 in user acceptability evaluations.
6. To architect and implement the integration API layer between the voice-NLP component and the three sister components of PaddyGuard AI, ensuring seamless handoff of disease predictions, confidence scores, and symptom metadata to the image-based verification and treatment recommendation modules.

3. METHODOLOGY

This research adopts an Experimental Research approach combined with Design Science Research (DSR) methodology, as it involves both the design and construction of novel AI artefacts and the empirical evaluation of those artefacts against measurable performance benchmarks.

3.1 Research Approach

The Design Science Research methodology (DSRM) by Peffers et al. provides the overarching framework, guiding the iterative cycle of problem identification, objective definition, artefact design, development, demonstration, and evaluation. This is complemented by an experimental ML research approach for the comparative evaluation of NLP classification models, following standard supervised learning protocols with train/validation/test splits.

3.2 Data Collection Strategy

Primary Dataset — Sinhala Voice Symptom Corpus:

A domain-specific dataset will be constructed through three channels: (1) structured data collection sessions with paddy farmers across three districts (Polonnaruwa, Kurunegala, Anuradhapura) in collaboration with the Department of Agriculture of Sri Lanka; (2) crowdsourcing via university volunteers who are native Sinhala speakers reading curated symptom descriptions; and (3) data augmentation using audio perturbation techniques (speed variation, background noise injection, pitch shifting) to improve model robustness. Each utterance will be labelled with one of four disease/health categories by agricultural domain experts.

Secondary Dataset — Text Symptom Corpus:

Transcribed symptom descriptions, agricultural extension literature in Sinhala, and translated content from established English-language rice disease databases (including IRRI's Rice Knowledge Bank and PlantVillage descriptions) will form a parallel text corpus for NLP model training.

Ethical Considerations: All human participants will provide informed consent. Data will be anonymised prior to model training. The study will be conducted following SLIIT Research Ethics Committee guidelines.

3.3 Technical Implementation Pipeline

Stage 1 — ASR Module (Whisper)

OpenAI Whisper (whisper-large-v3) will be deployed as the primary ASR engine. The model will be fine-tuned on the collected Sinhala agricultural speech corpus using a learning rate scheduler with warmup steps and CTC loss. Evaluation will be performed on a held-out 20% test split using Word Error Rate (WER) and Character Error Rate (CER) as primary metrics.

Stage 2 — Text Pre-processing and Symptom Extraction

Transcribed Sinhala text will undergo: (1) Unicode normalisation for Sinhala script; (2) stopword removal using a domain-adapted Sinhala stopword list; (3) Named Entity Recognition (NER) for agricultural entity extraction (plant parts: leaf, stem, root; visual attributes: colour, texture, shape; severity: mild, severe). A custom Sinhala tokenizer will be developed using SentencePiece to handle the agglutinative morphology of the Sinhala language.

Stage 3 — Disease Classification Models

Three models will be implemented and compared:

- Baseline: TF-IDF (unigrams + bigrams, max 10,000 features) + Linear SVM with class-weight balancing.
- BERT Fine-tuned: ‘bert-base-multilingual-cased’ or a Sinhala-specific BERT variant, fine-tuned with a classification head (4 output classes), using AdamW optimiser, learning rate $2e-5$, 10 epochs.
- RoBERTa Fine-tuned: XLM-RoBERTa-base fine-tuned with identical hyperparameter configuration for cross-comparison.

Performance will be evaluated using Accuracy, Macro-F1, Precision, Recall, and Confusion Matrix analysis.

Stage 4 — Adaptive Dialogue Module

A decision-tree-guided dialogue framework will be implemented using a confidence threshold mechanism. When the primary classifier’s top-class confidence score falls below $\tau = 0.70$, the system will engage the farmer with a maximum of three adaptive follow-up questions (e.g., ‘Are the spots more on older leaves or newer leaves?’). Questions are drawn from a curated expert-

defined question bank mapped to discriminating symptom features for each disease class. Responses are re-processed through the NLP pipeline to refine the classification.

Stage 5 — Text-to-Speech Output

Synthesised Sinhala speech responses will be generated using Google Cloud TTS (Sinhala WaveNet voice) as the primary option, with Coqui TTS (open-source, deployable on-premise) as a fallback. Responses include the predicted disease name, confidence level, follow-up questions (if applicable), and a brief advisory message. Speech quality will be evaluated through a Mean Opinion Score (MOS) survey administered to 30 native Sinhala speakers.

3.4 Comparison of Existing Systems

Table 2: Comparison of Existing System

System / Research Work	Main Function	Input Type	Language Support	Advisory Support	Limitations
Plantix Mobile App [16]	AI-based plant disease detection for crops	Leaf images	Primarily English and limited regional languages	Provides treatment suggestions	Requires clear leaf images; not optimized for voice interaction; limited support for local languages such as Sinhala
PlantVillage / Nuru AI [17]	Deep learning system for plant disease identification	Leaf images	English	Provides diagnostic feedback	Focuses mainly on image-based detection; requires internet connectivity and trained models for specific crops
Agrio Crop Protection Platform [18]	Commercial crop monitoring and disease detection	Image input	English	Offers crop advisory	Paid platform; not tailored for small-scale farmers in developing regions

Research Work: CNN-based Rice Disease Detection (Mohanty et al.) [19]	Machine learning approach for plant disease classification	Image dataset	Not language-dependent	No advisory system	Research focused only on classification accuracy; not integrated into a practical farmer-facing system
Proposed System – PaddyGuard AI	Multimodal rice disease and pest diagnosis with advisory system	Voice input + Leaf image	Sinhala language support	Knowledge-based treatment recommendations	Designed specifically for small-scale farmers with low digital literacy

Several existing systems have been developed for plant disease detection using artificial intelligence. Applications such as Plantix and PlantVillage Nuru provide crop disease diagnosis based on leaf images captured using mobile devices [16], [17]. Research studies have also demonstrated the effectiveness of deep learning techniques for image-based plant disease classification using large datasets such as PlantVillage [18]. However, most existing solutions rely heavily on image inputs and do not support voice-based interaction or local language interfaces for farmers. This limitation highlights the need for multimodal systems that integrate speech recognition, natural language processing, and image analysis to provide more accessible agricultural decision support tools.

3.5 Validation Strategy

Model performance will be validated using a stratified 70/15/15 train/validation/test split. Cross-validation (5-fold) will be used to ensure generalisability. User acceptance testing will involve structured trials with 20 farmers across two districts. The integration API will be validated through end-to-end system tests covering all four inter-component handoff scenarios.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall PaddyGuard AI System Overview

PaddyGuard AI is a mobile-responsive web application built on a React.js (Vite) frontend communicating with a Python FastAPI backend. The system is deployed on cloud infrastructure and integrates four AI-powered components through a central orchestration API layer. The farmer interacts exclusively through a single, intuitive interface - providing voice input or uploading images - while the backend intelligently routes data to the appropriate component services.

The complete system workflow proceeds as follows: (1) A farmer visits the PaddyGuard AI web app on any mobile browser and selects the voice input option; (2) The Sinhala voice recording is transmitted to the Voice-NLP Component; (3) The component returns a disease prediction and confidence score; (4) If confidence is sufficient (≥ 0.70) and no images are uploaded, treatment recommendations are generated by Component 3; (5) If confidence is low or the farmer voluntarily uploads leaf/pest images, Component 2 (disease image classifier) and Component 4 (pest detector) process the images; (6) All diagnostic outputs are consolidated by Component 3's RAG chatbot to generate a comprehensive, personalized treatment plan; (7) The final response is delivered in Sinhala text and synthesized speech.

High-Level System Architecture of PaddyGuard AI: Multimodal Intelligent System for Rice Disease and Pest Management

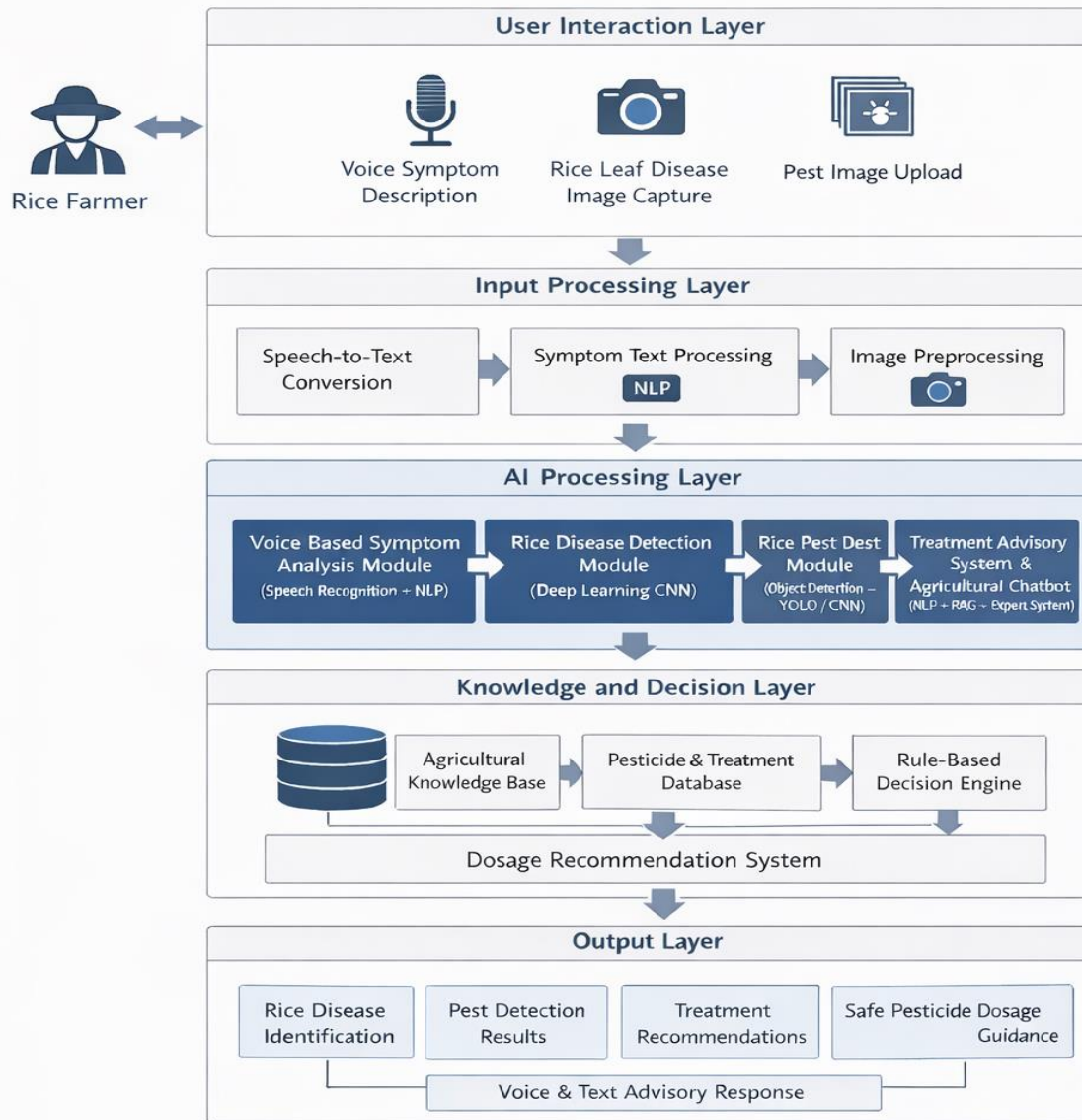


Figure 1: High Level Diagram

4.2 Detailed Architecture (Voice-NLP Pipeline)

The voice-NLP pipeline for this component is structured as five sequential processing stages

Table 3: NLP pipeline stages

Stage	Module	Technology	Output
1	Voice Capture & Transmission	Web Audio API / MediaRecorder	WAV/MP3 audio stream
2	Automatic Speech Recognition	OpenAI Whisper (large-v3)	Sinhala text transcript
3	Text Pre-processing & NER	SentencePiece, custom NER	Normalised symptom tokens
4	Disease Classification	BERT / RoBERTa fine-tuned	Class label + confidence score
5	Adaptive Dialogue / TTS Output	Decision tree + Google TTS	Sinhala spoken diagnosis

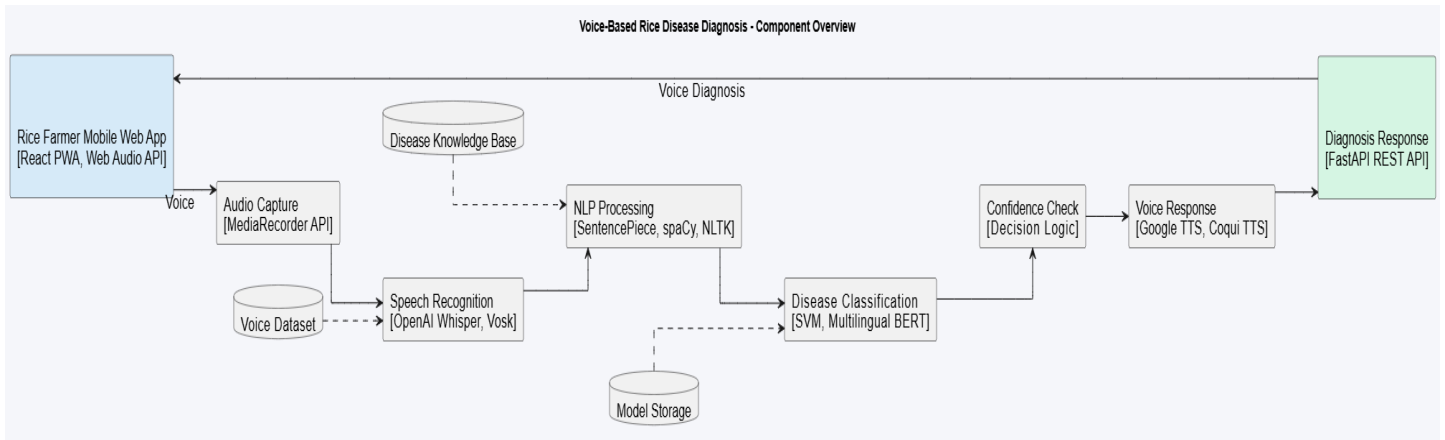


Figure 2: Voice Based Rise Disease Diagnosis Component High-Level Digram

4.4 Data Flow Diagram

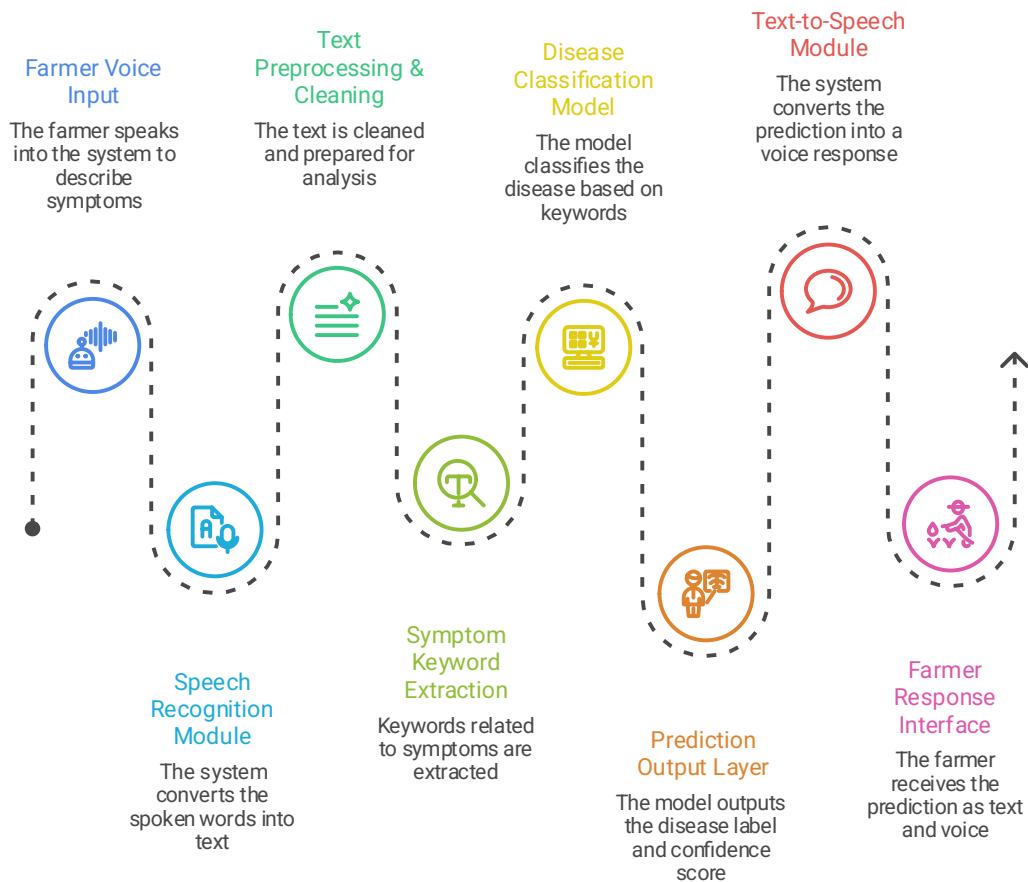


Figure 3: Voice Based Rise Disease Diagnosis Component Data Flow

4.5 Scalability and Security Considerations

The system is designed for horizontal scalability using containerised microservices (Docker + Kubernetes). Voice data is processed ephemeraly with no long-term retention of raw audio recordings. All API communications are secured using HTTPS/TLS 1.3. Farmer session data is stored in encrypted PostgreSQL databases. The system is designed to function on low-bandwidth connections (minimum 2G/EDGE) by compressing audio to 8kHz mono before transmission, ensuring usability in rural settings with limited connectivity.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary Stakeholders: Smallholder paddy farmers in Sri Lanka (low-literacy, Sinhala-speaking, rural).
- Secondary Stakeholders: Agricultural extension officers and agronomists who may use the system for field advisory support.
- Tertiary Stakeholders: Department of Agriculture Sri Lanka, agricultural input suppliers (pesticide/fertiliser companies), crop insurance providers.

5.2 Functional Requirements

FR01 — The system shall accept Sinhala voice input from a web browser microphone without requiring any additional app installation.

FR02 — The system shall transcribe Sinhala agricultural speech with a Word Error Rate (WER) of no greater than 15% under standard field noise conditions.

FR03 — The system shall classify the transcribed symptom description into one of four categories (Bacterial Blight, Rice Blast, Brown Spot, Healthy) with a minimum classification accuracy of 85%.

FR04 — The system shall display the predicted disease name in both Sinhala and English, along with a confidence indicator (Low / Medium / High).

FR05 — When confidence is below 0.70, the system shall present a maximum of three voice-based adaptive follow-up questions to the farmer.

FR06 — The system shall return all diagnostic outputs as synthesised Sinhala speech via TTS.

FR07 — The system shall detect pest-related keywords in voice input and trigger the pest detection component (Component 4) automatically.

FR08 — The system shall transmit diagnostic results (disease class, confidence, symptom metadata) to the treatment recommendation module (Component 3) via internal API.

FR09 — The system shall accept optional image uploads and pass them to Component 2 for multi-modal confirmation, merging the results with the voice-based prediction.

FR10 — The system shall function on mobile browsers (Chrome Android, Safari iOS) and display a responsive UI suitable for small screens.

5.3 Non-Functional Requirements

NFR01 — Performance: Voice-to-prediction latency shall not exceed 8 seconds on a standard 4G mobile connection.

NFR02 — Usability: The interface shall follow a voice-first, minimal-tap design requiring no more than 2 taps to initiate a voice diagnosis session.

NFR03 — Availability: The system shall maintain 99.5% uptime during peak paddy cultivation seasons.

NFR04 — Security: No raw audio data shall be stored beyond the processing session; all personally identifiable information shall be encrypted at rest and in transit.

NFR05 — Accessibility: The UI shall achieve WCAG 2.1 AA accessibility compliance and support low-bandwidth operation at 2G/EDGE network speeds.

NFR06 — Scalability: The system shall support a minimum of 500 concurrent users without performance degradation.

NFR07 — Accuracy: Overall system diagnostic accuracy (voice-only pathway) shall achieve at least 85% across all four disease categories on the test dataset.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is Sri Lanka's 1.8 million smallholder paddy farming households, with a reachable addressable market of approximately 500,000 digitally-accessible farming households. The secondary market includes agricultural cooperatives, government extension services (the Department of Agriculture's 1,500+ extension officers), and regional rice exporters seeking quality assurance tools.

Primary Persona — 'Farmer Sunil': Male, 45 years, paddy farmer in Polonnaruwa, Sinhala-only speaker, owns a basic Android smartphone, uses WhatsApp for family communication, has no prior experience with agricultural apps, experiences disease losses of 20–30% annually due to delayed diagnosis.

6.2 Value Proposition

- Immediate, in-field diagnosis in the farmer's native Sinhala language without requiring camera access.
- Reduction of crop loss by enabling early intervention - research shows early treatment reduces disease-related yield loss by 60–70%.
- No literacy requirement: entirely voice-driven interaction model.
- Free at point of use for farmers; monetisation through B2G and B2B channels.

6.3 Revenue Model

- Government (B2G): Licensing to the Department of Agriculture of Sri Lanka as an extension service tool - projected licensing fee LKR 5M–15M per year.
- Agri-input Companies (B2B): White-label API integration for pesticide and fertiliser companies to reach farmers at point of diagnosis.
- Development Partners and NGOs: Grant funding from FAO, CGIAR, and ADB for deployment in other South and Southeast Asian rice-growing nations.
- Premium Features (SaaS): Advanced analytics dashboard for cooperatives and extension officers - LKR 2,500/month per organisation.

6.4 Competitive Advantage

PaddyGuard AI is the only known system offering Sinhala voice-based rice disease diagnosis with multimodal confirmation. Competitors such as Plantix (German, image-only, no Sinhala) and AgroStar (India-focused, limited Sinhala support) do not address the specific linguistic, literacy, and connectivity challenges of Sri Lankan paddy farming communities.

6.5 Intellectual Property

The custom Sinhala agricultural speech corpus, the fine-tuned Whisper ASR model, the fine-tuned BERT/RoBERTa disease classifier, and the adaptive dialogue question bank constitute the core IP of this research. These will be registered under SLIIT's research commercialisation framework, with open publication of the dataset to support the broader research community under a Creative Commons Attribution (CC BY) licence.

7. BUDGET AND JUSTIFICATION

7.1 Estimated Project Budget

This section presents the estimated budget required for the development and evaluation of the proposed **PaddyGuard AI system**. Since this research is conducted as part of an undergraduate academic project, the overall cost is minimized by utilizing open-source software tools, existing personal computing resources, and free or low-cost cloud platforms wherever possible.

However, certain essential resources are required for machine learning experimentation, dataset storage, model training, and prototype deployment. Therefore, a limited budget allocation is considered for hardware upgrades, cloud computing resources, software tools, and documentation preparation. These resources ensure that the project can be implemented efficiently while maintaining technical feasibility within an academic environment.

7.2 Hardware Costs

The research will primarily be conducted using an existing personal laptop owned by the researcher. Therefore, major hardware investments are not required. However, small hardware upgrades may be necessary to support dataset storage and machine learning experiments.

For example, pest image datasets and trained deep learning models may require additional storage space. Furthermore, an optional RAM upgrade may improve model training and data processing performance.

Table 4: Hardware cost

Hardware	Purpose	Estimated Cost (LKR)
External Storage (1TB HDD / SSD)	Used to store pest image datasets, trained models, and backup research data securely	5,000
RAM Upgrade (Optional – 8GB/16GB)	Improves system performance during machine learning model training and dataset preprocessing	12,000
Total Cost		17,000

Using an existing personal laptop significantly reduces the overall hardware cost while still providing sufficient computational resources for experimentation and prototype development.

7.3 Software Licenses

The development of the PaddyGuard AI system mainly relies on open-source software technologies widely used in artificial intelligence and computer vision research. These tools provide powerful development capabilities without requiring expensive licensing fees.

Table 5: Software License

Software	Purpose	License Type	Cost (LKR)
Python 3.x	Programming language used for AI model development and data processing	Open Source	0
PyTorch / TensorFlow	Deep learning frameworks used for training pest detection models	Open Source	0
OpenCV	Image processing and computer vision library used for pest detection tasks	Open Source	0
NumPy / Pandas	Data processing and numerical computation libraries	Open Source	0
VS Code	Integrated development environment used for coding and debugging	Free	0
Label Studio / Annotation Tools	Used to label pest image datasets for training the detection model	Open Source	0
TOTAL			0

The use of open-source technologies significantly reduces software licensing expenses while still providing access to advanced machine learning frameworks required for this research.

7.4 Cloud Services

Cloud computing services may be used to support large-scale model training, dataset storage, and system experimentation. These services provide GPU-accelerated computing environments that significantly speed up deep learning model training.

Cloud platforms also support scalable storage solutions and allow deployment of prototype APIs for testing the AI-based pest detection system.

Table 6: Cloud Service

Service	Usage	Plan	Cost (LKR)
Google Colab Pro	GPU-accelerated training environment for deep learning models	Monthly subscription	15,000
Cloud Storage (AWS S3 / Google Cloud Storage/ Azura)	Storage for pest image datasets, trained models, and backup files	Pay-as-you-go	6,000
Cloud Compute (AWS / Azure / GCP)	Hosting AI model APIs and prototype backend services	Pay-as-you-go	5,000
Domain & SSL Certificate	Secure HTTPS API endpoint for prototype deployment	Annual	3,500
TOTAL			29,500

Cloud-based GPU resources significantly reduce the need for expensive hardware upgrades while enabling efficient training of deep learning models used for pest detection.

7.5 Human Effort Cost

Since this research project is conducted as part of an undergraduate academic program, no direct labor costs are included. However, a small budget allocation is considered for essential academic and research-related activities. This includes printing and binding costs for proposal reports, progress reports, and final thesis documentation required for submission and evaluation. In addition, limited funding is allocated for field data collection activities, which involve visiting paddy cultivation areas and interacting with farmers to collect voice samples and validate the voice-based system under real agricultural conditions. These activities support the creation of a realistic dataset and help ensure that the developed system functions effectively in practical farming environments.

Table 7: Human Effort Cost

Role	Task	Cost (LKR)
Printing & Binding	Printing and binding of proposal reports, progress reports, and final thesis documentation	3,000
Field Data Collection Travel	Transportation expenses for visiting paddy cultivation areas to collect voice samples and conduct farmer interviews required for dataset preparation and validation.	10,000
TOTAL		13,000

7.6 Justification of Expenses

Table 8: Justification of Expenses

Category	Total Cost (LKR)
Hardware Costs	17,000
Software Licenses	0
Cloud Services	29,500
Human Effort Cost	13,000
GRAND TOTAL	59,500 LKR

The estimated budget reflects the essential resources required to successfully develop and evaluate the **PaddyGuard AI system**. Hardware upgrades such as additional storage and memory improve the capability to manage large pest image datasets and machine learning model outputs.

Cloud-based GPU services enable efficient training of deep learning models such as **ResNet50, EfficientNet, and YOLOv8** without requiring expensive high-performance hardware. Cloud storage solutions also support secure dataset management and prototype deployment.

Furthermore, the use of open-source technologies significantly reduces software licensing costs while still providing powerful tools for artificial intelligence development. Overall, the proposed budget ensures that the research project remains **cost-effective, technically feasible, and suitable for an academic research environment** while supporting the successful development of the AI-based pest detection system.

8. WORK BREAKDOWN STRUCTURE (WBS)

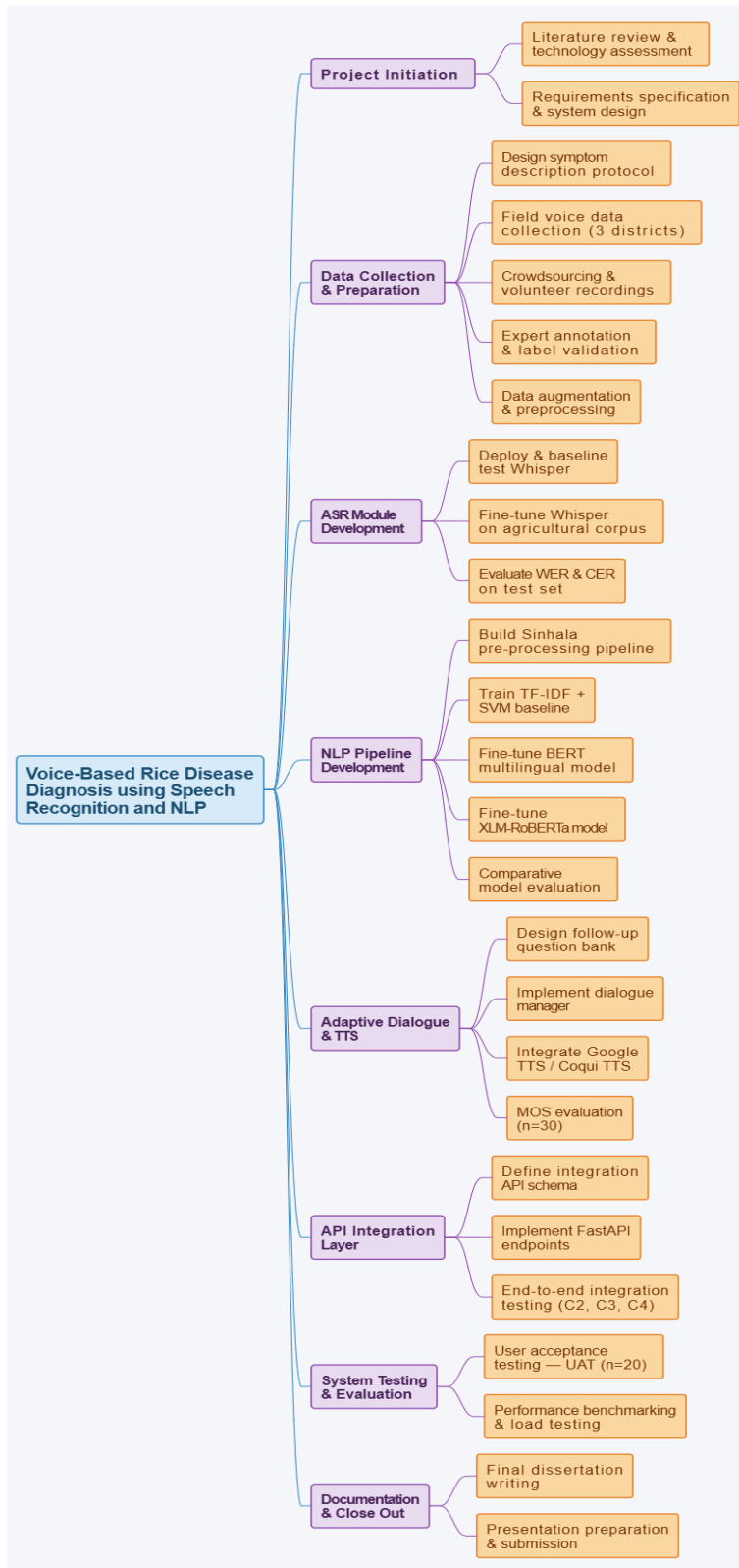


Figure 4: WBS

9. GANTT CHART

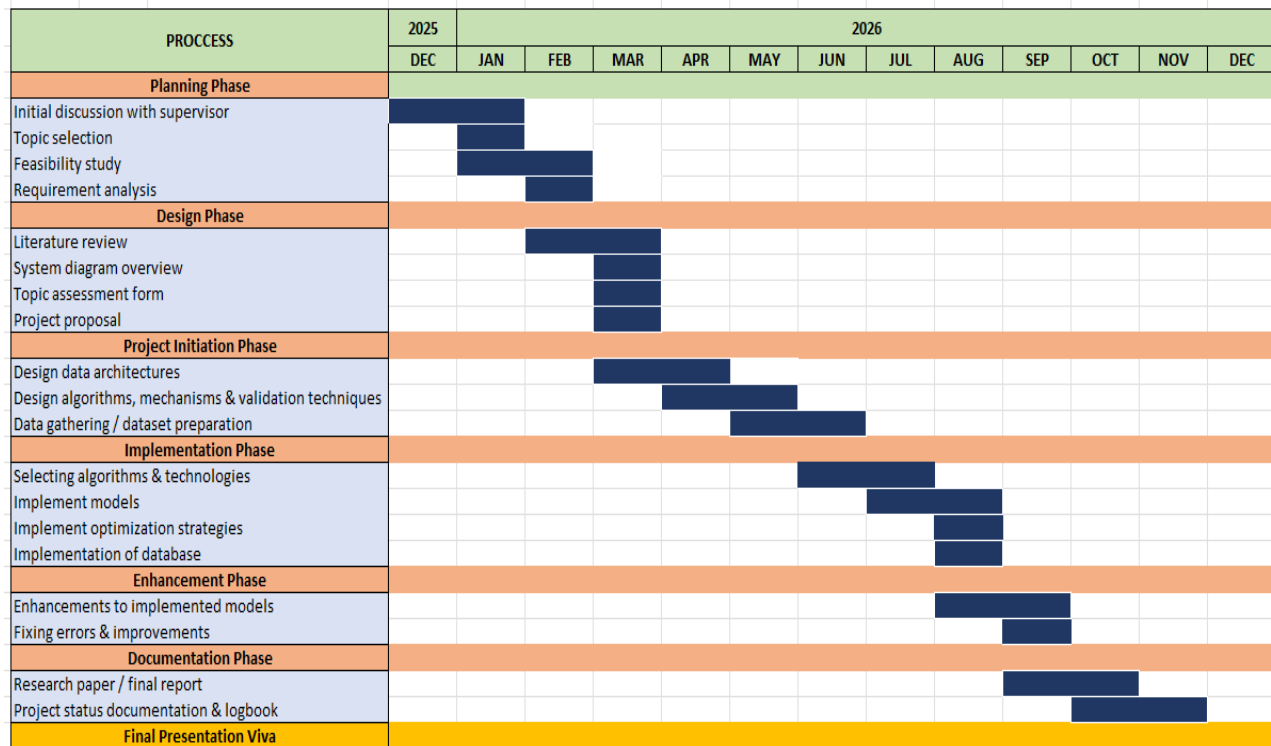


Figure 5: Gantt Chart

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11. APPENDICES

11.1 Preliminary UI mockups

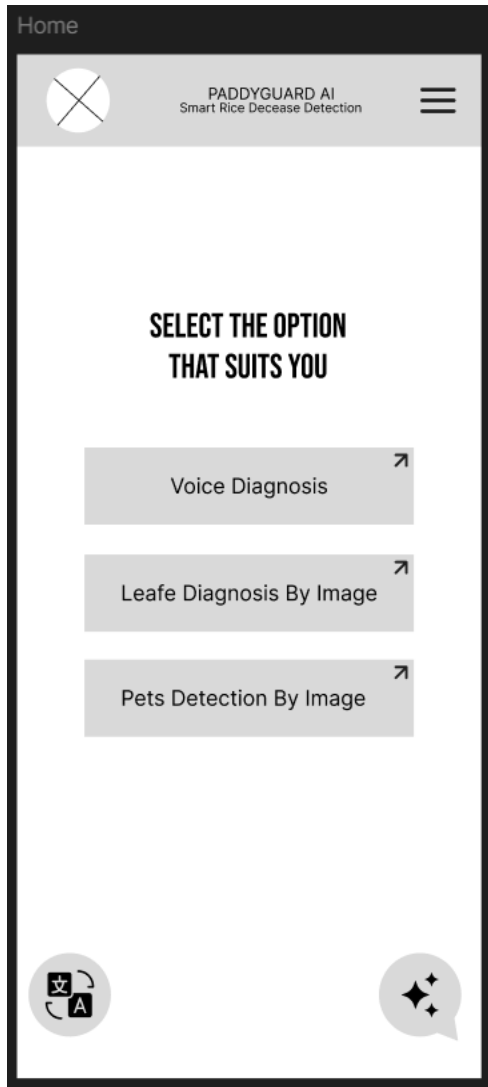


Figure 7: Home Page

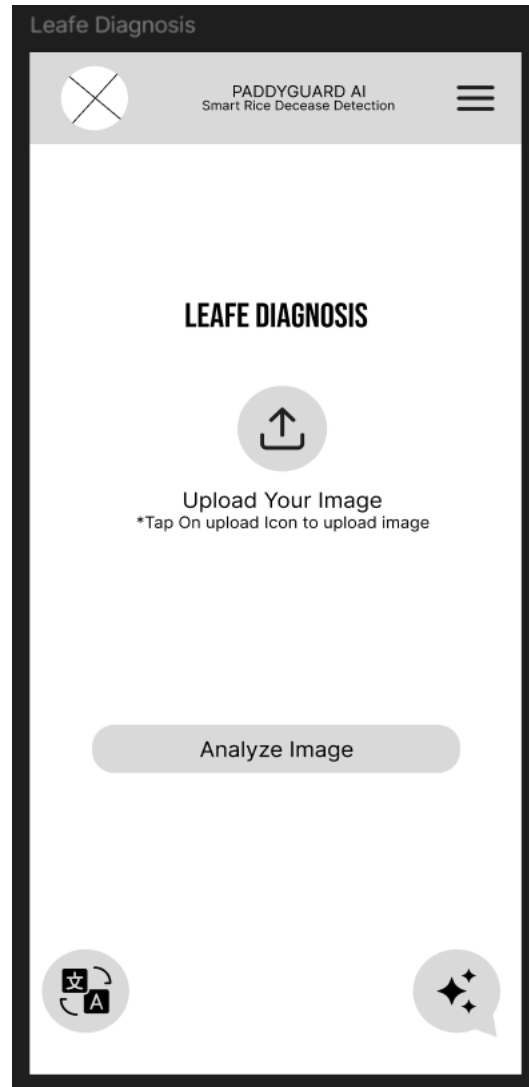


Figure 6: Leaf Diagnosis

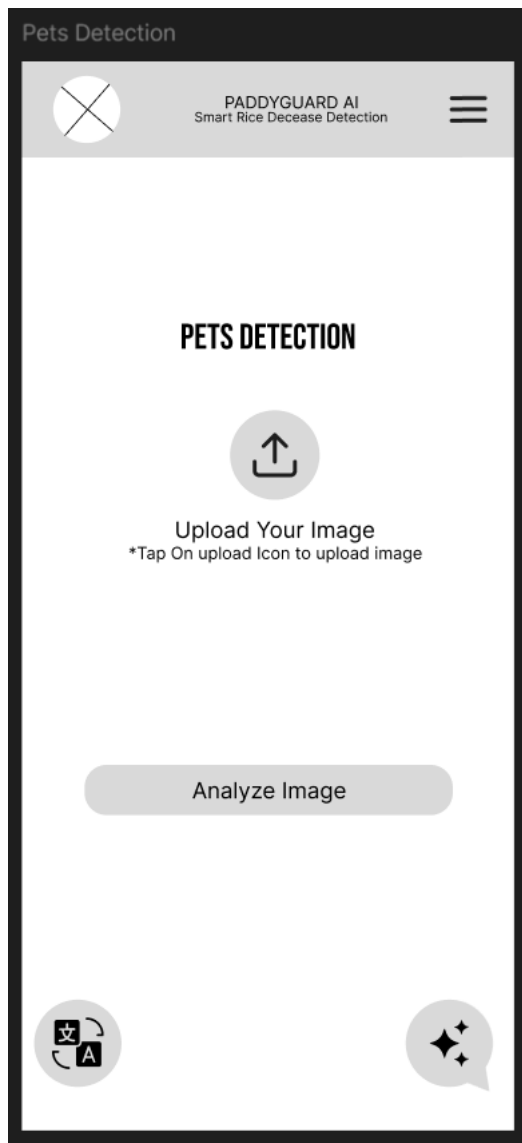


Figure 9: Pets Detection

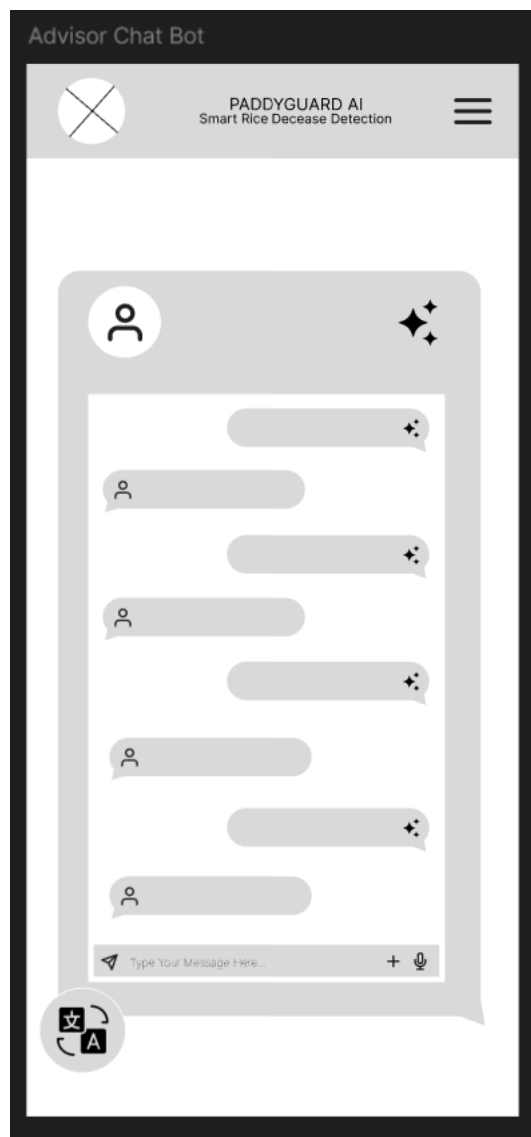


Figure 8: Chat Bot

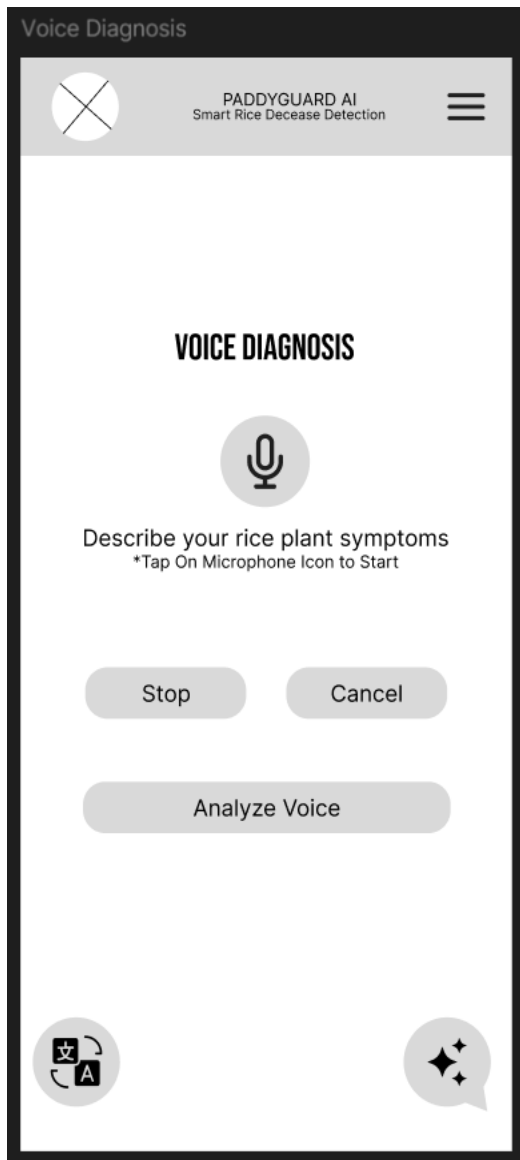


Figure 11: Voice Diagnosis

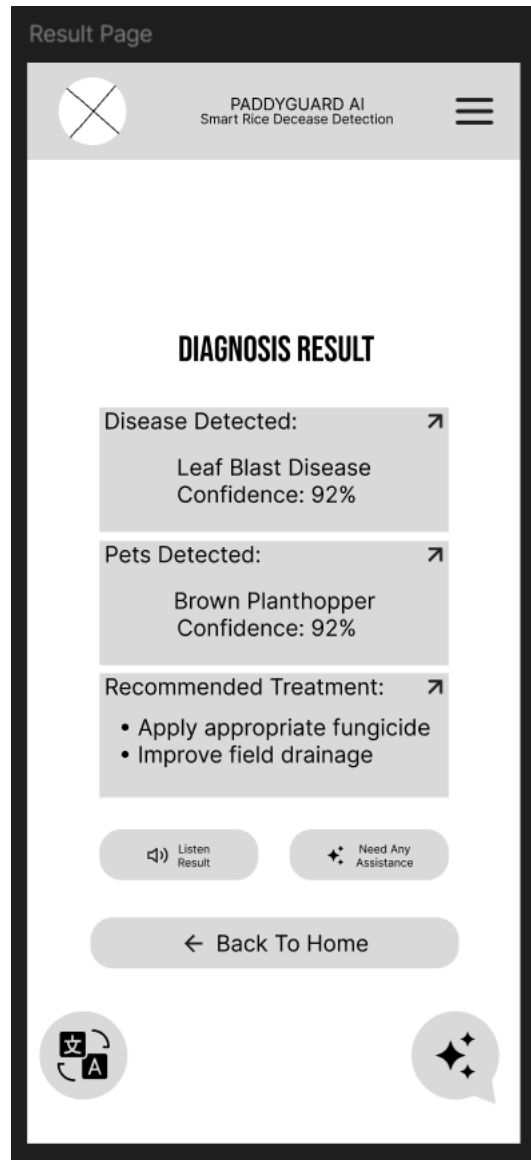


Figure 10: Result Page